



**The voice agent that runs
a real technical interview —
and coaches you **while** you talk.**

Built on AssemblyAI Universal-Streaming

THE PROBLEM

**Everyone trains for the interview.
Almost nobody trains for **explaining** the answer.**

01

LeetCode grades code

It optimises the one part of an interview you are already good at, and is silent on the part that actually fails people.

02

Feedback arrives too late

"Clearer communication" lands in a rejection email weeks later, with no idea which sentence cost you the offer.

03

Mock interviews don't scale

A human interviewer cannot listen for structure, complexity and filler words at the same time. So they don't.

THE INSIGHT

A strong answer has a **shape** — and you can score it live, mid-sentence.

Every credible algorithmic explanation covers the same five steps. Once you name them, you can watch a transcript fill in as it happens — and tell the speaker, in real time, which step they skipped.

01

Restate

Confirm the actual task in your own words.

02

Approach

Commit to a direction before exploring it.

03

Data structure

Name the structure and why it fits.

04

Complexity

Time *and* space, in big O.

05

Edge cases

Empty input, duplicates, the boundaries.

WHAT IT DOES

You talk. The rubric fills in.

- **Asks the question out loud** and opens the mic without a click.
- **Scores the live transcript** against the five steps, from partial results.
- **Interrupts a ramble** when you talk past the limit without analysing anything.
- **Probes instead of grading** when an answer is too thin to score.

ANSWER METER — LIVE

Restate		hit
Approach		hit
Data structure		hit
Complexity		...
Edge cases		—

Complexity is half-credited and edge cases are untouched. The interviewer is about to ask for exactly that.

ARCHITECTURE

Audio goes straight to AssemblyAI. The server never touches it.

Browser

AudioWorklet converts mic float samples to PCM16 @ 16 kHz, 100 ms frames, and reads RMS for the live waveform.

→
WebSocket
raw PCM

AssemblyAI

Universal-Streaming `universal-3-6-pro`. Partial turns, final turns, word timestamps, end-of-turn detection.

←
transcript
+ timings

Server

tRPC only: mints single-use streaming tokens and grades answers. No audio is proxied, buffered or stored.

One process, no media server

Express serves the built client and answers two tRPC calls. Deployable anywhere Node runs.

The account key never ships

The browser requests a 60-second, single-use token per turn. The key stays server-side.

Features that made the coaching possible

FEATURE	WHY THE PRODUCT NEEDS IT	WHERE
Universal-Streaming	Sub-second partial transcripts, so the rubric can update mid-sentence	<code>assemblyaiStream.ts</code>
End-of-turn detection	The server decides when the candidate stopped talking — no silence timers, no clipped answers	<code>assemblyaiStream.ts</code>
Temporary tokens	Single-use per turn, so the account key is never exposed to the browser	<code>_core/assemblyai.ts</code>
Word-level timestamps	Real WPM, longest pause and filler rate rather than a guess	<code>shared/rubric.ts</code>
Keyterms per question	Forces "Big O", "two pointers" and "hash map" to transcribe correctly	<code>questionBank.ts</code>
Agent context	Biases the model toward interview vocabulary and spoken notation	<code>assemblyaiStream.ts</code>
Inactivity timeout	Bounds a stalled turn instead of billing an open socket forever	<code>assemblyaiStream.ts</code>

Three bugs that would have killed a live demo

BUG 01

The cleanup that killed every session

The voice hook returned a new object each render, so the unmount effect re-ran constantly and tore down the live microphone. Effects now depend on stable callbacks.

BUG 02

Turn two onward had no audio

AudioWorklet registration is per AudioContext, not global. Our "already loaded" cache meant only the first interview turn could capture audio.

BUG 03

A watchdog that could never fire

The ramble timer depended on the transcript, so each new word restarted it. It now reads a ref, and interrupts long unstructured answers as designed.

Found by running the shipped AudioWorklet source against a stubbed Web Audio global in the test suite — 26 tests covering the rubric, delivery maths and audio resampling.

Honest about what it is

- **Coaching at the moment of the mistake** is the entire product. Feedback after the fact is what candidates already have.
- **Works without a microphone.** Skip and typed-answer paths mean the grading engine is never gated on speech.
- **Degrades honestly.** No LLM key falls back to a heuristic grader instead of pretending.
- **No audio is stored.** Transcripts live in memory for the length of the session.
- **It's practice, and it says so.** The interviewer never claims to be a recruiter or promises an outcome.
- **Next:** two-way voice, resume-anchored questions, and a rubric that adapts to your role.



Stop practising the code. Practise the **answer**.

Live demo, source and the AssemblyAI integration are in the repository.

React + TypeScript

AssemblyAI Universal-Streaming

tRPC

AudioWorklet

26 tests passing